

Cybersecurity & Backend Development Notes

Study Notes

October 5, 2026

Day 7 — To Understand

1. Data Flow Diagram (DFD)
 2. Who to integrate — payment gateway
 3. How to secure the application
 4. Where to deploy — how to deploy the application
 5. What is meant by web service / web application
 6. What is API and API gateway
-

Application Layer Architecture

1. 3-Tier Architecture

- a. **Presentation layer** — UI / Frontend → Business logic, APIs / services
- b. **Application layer**
- c. **Data layer** (Database) — a and b together are called the **backend layer**

The **presentation layer** designs the web application or mobile application and display. The **application layer** handles web, mobile, and other clients.

Frontend Tools

- HTML
- CSS
- JS (Next.js, Vue.js, React.js) — these are frontend frameworks/libraries

Difference Between Frontend Tools (Languages) and Frontend Frameworks

#	Frontend Tools (Languages)	Frontend Frameworks
1	Core building-block languages of the web	Pre-built libraries/structures built on top of those languages
2	Example: HTML, CSS, JavaScript	Example: React.js, Vue.js, Next.js, Angular
3	You write everything from scratch	Provide ready-made components and patterns to speed up development
4	No built-in structure or rules — total freedom	Enforce a specific project structure and coding pattern
5	Browser understands them natively (no compilation needed)	Often need a build step/compiler (e.g., JSX → JS, Webpack, Vite)
6	Handle only presentation (HTML), styling (CSS), and basic logic (JS)	Handle state management, routing, component reuse, and app-wide logic

7	Best for small, simple static pages	Best for large, complex, interactive single-page applications (SPAs)
8	No dependency management needed	Installed via package managers (npm/yarn) and have many dependencies
9	Updating the UI means manually changing the DOM	Automatically re-render the UI when data/state changes (Virtual DOM)
10	Example: A plain HTML+CSS+JS portfolio page	Example: A React.js e-commerce site with cart, login, and dynamic product listings

Simple example: - *Using frontend tools only:* `<button onclick="count++; document.getElementById('c').innerText=count">Click</button>` — you manually update the DOM every time.
- *Using a frontend framework (React):* `<button onClick={() => setCount(count + 1)}>Click</button>` — React automatically re-renders the count on the page whenever state changes, without you touching the DOM directly.

Backend Layers

In backend layers, there are two types of layers: 1. Application layer (business logic) 2. Data layer

How to choose which language to use depends on the need.

Remember — List of Backend Languages

1. Java
2. Java Spring Boot, Microservices
3. Go (Golang)
4. Python
5. PHP
6. C#
7. Rust

Backend Framework Tools

1. Python (Django, Flask)
2. Node.js
3. Java Spring Boot
4. ASP.NET Core
5. Go — Gin
6. PHP — Laravel (*the most used*)

Backend Framework vs Backend Language

What does a backend framework feature?

1. Routing
2. REST API
3. Business logic

4. Database integration
5. ORM
6. Authentication & session management
7. Authorization
8. (blank)
9. Input validation
10. Error handling
11. Middleware
12. Security
13. Logging
14. Caching
15. File uploads
16. WebSockets

4 Types of Databases

1. Hierarchical database
2. Network database
3. RDBMS
4. NoSQL database

What is a database and database management?

Under **RDBMS**, there are two types: 1. SQL (Excel-like) 2. NoSQL (MS Word-like)

SQL Tools

1. PostgreSQL
2. MySQL
3. IBM DB2
4. *(more to add)*

Types of NoSQL Databases

Type	Data Structure	Popular Tools	Common Use
Document	JSON/BSON documents	MongoDB, CouchDB	Web/mobile apps, content
Key-Value	Key-value pairs	Redis, DynamoDB	Caching, sessions, high-speed lookup
Wide-Column	Rows & column families	Cassandra, HBase, ScyllaDB	Large-scale, write-heavy workloads
Graph	Nodes + relationships	Neo4j, Amazon Neptune	Relationships & network analysis

Type: Multi-node

Application — Cybersecurity Practical Toolset

Requirements

1. Domain
2. Linux server

3. Frontend UI layer
 4. Deployed UI layer — server (Linux server, AWS cloud)
-

OSINT (Open-Source Intelligence)

OSINT means collecting and analyzing information from publicly accessible sources, such as:

- Websites
- Search engines
- Social media
- Public email information
- WHOIS and domain records
- IP addresses
- Public documents
- Company information
- GitHub / public code repositories
- Blogs and forums

An **OSINT framework** is essentially a catalogued directory / map of OSINT tools and resources.

Think of it like: I have a target domain — what publicly available information can I bring in, and which tools do I use?

| Hydra

Hydra is a network login/password auditing tool.

Hydra automates authentication attempts against network services using a username and a list of candidate passwords.

It supports many protocols, including: SSH, FTP, HTTP, HTTPS, SMTP, POP3, IMAP, SMB, RDP, MySQL, PostgreSQL, Telnet, LDAP.

Nmap (Network Mapper)

Nmap is one of the most important tools for network security and ethical hacking.

Nmap is a network discovery and security auditing tool used to find live hosts, open ports, running services, and sometimes operating system information on a network.

| What Nmap Can Do

- **Host discovery** – Find active devices
- **Port scanning** – Identify open, closed, or filtered ports
- **Service detection** – Determine what service is running on a port
- **Version detection** – Identify service versions
- **OS detection** – Attempt to identify the operating system
- **Network inventory** – Help administrators understand their network
- **Security auditing** – Find exposed services that may need protection

Basic Example

```
nmap 192.168.1.10
```

This scans the specified host for commonly used ports.

⚠ For cybersecurity training, use Nmap only against systems you own or have explicit permission to test.

Example Port Mappings

- Port 22 → SSH
- Port 80 → HTTP
- Port 443 → HTTPS

What is a Port?

A port is a logical communication endpoint used by a computer to identify which application/service should receive network traffic.

Port	Service
22	SSH
53	DNS
80	HTTP
443	HTTPS
3306	MySQL
3389	RDP

Port Number Range

The TCP/UDP port number range is **0–65535** (65,536 total possible ports).

Main Port Number Categories

Range	Category	Description
0–1023	Well-known ports	Common standard services
1024–49151	Registered ports	Registered for applications/vendor services
49152–65535	Dynamic / Private (ephemeral) ports	Usually used temporarily for client-side connections

Common Protocols by Port

Port	Protocol	Purpose
20, 21	FTP	FTP data / FTP control
22	SSH	Secure remote administration
23	Telnet	Remote terminal (unencrypted)
25	SMTP	Mail transfer
53	DNS	Domain Name System
67, 68	DHCP	DHCP server / DHCP client

69	TFTP	Trivial File Transfer
80	HTTP	Web
110	POP3	Email
123	NTP	Time
143	IMAP	Email
443	HTTPS	Secure web
445	SMB	Windows file sharing
389	LDAP	Directory services
3306	MySQL	Database
3389	RDP	Remote Desktop
5432	PostgreSQL	Database
6379	Redis	Database / cache
8080	HTTP alternate	Web (alt port)

Easy memory: - 0–1023 → Well-known - 1024–49151 → Registered - 49152–65535 → Dynamic/Private

Voltage vs Current

Voltage	Current
Electrical potential difference between two points	Flow of electric charge through a circuit
Provides the push that drives charge	Is the movement/flow of charge
Unit: Volt (V)	Unit: Ampere (A)
Measured using a voltmeter	Measured using an ammeter
Symbol: V	Symbol: I

Simple analogy: Think of water in a pipe — voltage is the water pressure pushing the water, current is the amount of water flowing through the pipe.

Ohm's Law: $V = I \times R$, where V = voltage, I = current, R = resistance.

How to Install Kali Linux on Windows 11 (via WSL)

1. **Enable WSL** — Open PowerShell as Administrator and run `wsl --install`. This enables the Windows Subsystem for Linux and installs the default Ubuntu distro. Restart your PC when prompted.
2. **Check WSL version** — After restart, open PowerShell and run `wsl --set-default-version 2` to make sure WSL2 is the default (Kali runs best on WSL2, not WSL1).
3. **Install Kali Linux from Microsoft Store** — Open the Microsoft Store app, search for "Kali Linux," and click Install. Alternatively, run `wsl --install -d kali-linux` from PowerShell.
4. **Launch Kali and set up user** — Open Kali Linux from the Start menu. On first launch it will initialize and ask you to create a UNIX username and password — this is your Kali login, separate from Windows.
5. **Update and upgrade Kali** — Inside the Kali terminal, run:

```
sudo apt update && sudo apt full-upgrade -y
```

6. **Install the full Kali toolset (optional)** — The Store image is minimal by default (no GUI tools like Nmap/Hydra preinstalled). Run:

```
sudo apt install kali-linux-large
```

or `kali-linux-default` for a smaller common set.

7. **(Optional) Enable GUI apps** — WSL2 on Windows 11 supports GUI Linux apps natively (WSLg). Tools with a GUI (like Wireshark) open as normal windows — no extra X server setup needed.

Alternative: For the full Kali desktop experience, download the Kali VirtualBox/VMware image from kali.org and run it as a full virtual machine instead — better for GUI-heavy tools, with slightly more setup and resource usage.

Question & Answer Log

Q: What is Nmap? A: Nmap (Network Mapper) is a network discovery and security auditing tool used to identify hosts, open ports, services, and sometimes operating-system information. It can do host discovery, port scanning, service detection, version detection, OS detection, network inventory, and security auditing. Basic example: `nmap 192.168.1.10` scans the specified host for commonly used ports. For cybersecurity training, use Nmap only against systems you own or have explicit permission to test.

Q: What is a port number? A: A port number is a 16-bit number (0–65535) used to identify a specific service or application on a device in a network. Examples: 22 → SSH, 53 → DNS, 80 → HTTP, 443 → HTTPS, 3306 → MySQL, 3389 → RDP. Simple analogy: IP address = house address, port number = room number. So 192.168.1.10:443 means the device 192.168.1.10 is being accessed through port 443, commonly used for HTTPS.

Q: Give me port number 0 to 65533 A: The TCP/UDP port number range is 0–65535, not 0–65533 — a total of 65,536 possible port numbers.

Q: What are the main port number categories? A: Port numbers 0–65535 are divided into 3 main categories: Well-Known Ports (0–1023) used by standard services like SSH, DNS, HTTP, HTTPS; Registered Ports (1024–49151) registered for specific applications like MySQL, PostgreSQL, HTTP alternate; and Dynamic/Private Ports (49152–65535) usually used temporarily for client-side connections.

Q: What is the difference between voltage and current? A: Voltage is the electrical potential difference between two points and provides the push that drives charge (unit: Volt, symbol V, measured with a voltmeter). Current is the flow of electric charge through a circuit — the movement of charge itself (unit: Ampere, symbol I, measured with an ammeter). Simple analogy: in a water pipe, voltage is the water pressure pushing the water, and current is the amount of water flowing through the pipe. Ohm's Law: $V = I \times R$.

Q: What is Hydra? A: Hydra is an open-source password-cracking / login-auditing tool used in cybersecurity to test the strength of authentication services. It can test services such as SSH, FTP, HTTP, SMB, and some database services by attempting many username/password combinations. In ethical hacking, it's used to test whether passwords are weak, audit authentication security, identify accounts vulnerable to password-guessing attacks, and demonstrate the importance of strong passwords and MFA. Use Hydra only on systems or accounts you own or have explicit permission to test.

Q: Explain Hydra with an example A: Suppose you have a practice SSH server in your own cybersecurity lab with username "student" and an unknown password. You create a small test list (password1, welcome123, admin123, correct-password). Hydra tests these credentials against the authorized lab account and reports if one works. Conceptually: Hydra → username + password list → login attempts → authentication response → weak password identified. Lesson: weak password → password guessing possible → use strong passwords + MFA + rate limiting/account lockout.

Q: How to install Kali Linux on Windows 11? A: The easiest method is via WSL (Windows Subsystem for Linux):

enable WSL (`wsl --install`), set WSL2 as default, install Kali Linux from the Microsoft Store (or `wsl --install -d kali-linux`), launch it and set up a UNIX user, then update/upgrade and optionally install the full toolset with `sudo apt install kali-linux-large` . WSL2 on Windows 11 also supports GUI Linux apps natively via WSLg. Alternatively, run Kali as a full VirtualBox/VMware virtual machine for the complete desktop experience.